

Dark Energy with Magnetized Strange Quark Matter in $f(R, L_m)$ Gravity

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Received August 26, 2024; revised October 18, 2024; accepted December 10, 2024

Abstract—We investigate a Bianchi-type VI cosmological model that integrates magnetized strange quark matter (MSQM) into the context of $f(R, L_m)$ gravity, where R denotes the Ricci scalar, and L_m signifies the matter Lagrangian. The model examines a nonlinear functional expression of $f(R, L_m) = R/2 + \lambda R^2 + \alpha L_m$, resolved by a linearly variable deceleration parameter. We derive cosmological parameters from redshift, including the Hubble parameter, and compare the model predictions with the empirical Hubble dataset values. We also examine the energy conditions within the model in detail, particularly, the effects of dark energy. This paper offers an extensive examination of the physical and geometrical characteristics of the universe while illustrating the significance of strange quark matter, magnetic fields, and dark energy in modified gravity theories. Some of the most important things we found were new information on how the deceleration parameter, anisotropy, and shear scalar behave. This helps us to understand how the universe is expanding in modified theories of gravity.

DOI: 10.1134/S0202289325700124

1. INTRODUCTION

Many scientific investigations have been conducted over the last few decades in order to comprehend the mysterious behavior of the universe. From early inflation to late-time acceleration, black holes, wormholes, dark energy (DE), and gravitational waves, the nature of the universe has been explored throughout its history. In terms of cosmology, the universe is considered both isotropic and homogeneous on a wide scale. In 1992, the Cosmic Background Explorer identified a tiny anisotropy in the large-scale cosmic microwave background. More information came from the Ballon Observations of Millimetric Extragalactic Radiation and Geophysics (BOOMERang) [2], the Cosmic Background Imager [3], the Wilkinson Microwave Anisotropy Probe (WMAP) [4], and the Planck collaborations [5]. Furthermore, exciting breakthroughs in the discipline of cosmology were achieved through observational outcomes [6, 7]. These investigations support the idea that the universe is expanding faster than ever. Recent research indicates that the universe expands at varying rates and in diverse directions [8]. However, observable data attempts to explain the existence

of slight variations in the intensities of microwaves emanating from various axes. This is why Bianchi's cosmologies are the space-time that can adequately represent anisotropic and homogenous geometries. The literature has a number of studies on such Bianchi cosmologies using various modified gravity frameworks [9–15].

Furthermore, one of the greatest ways to explain the accelerated expansion of the universe is through modified theories of gravity [16–18]. A few of them are commonly referred to as $f(R)$ gravity, $f(Q)$ gravity, $f(R, T)$ gravity, and $f(R, L_m)$ gravity [19–23]. The $f(R)$ gravity was extended, and it was suggested to explicitly couple the matter Lagrangian density L_m with the generic function $f(R)$ [24]. We extended this model [25] to include the case of arbitrary couplings in both matter and geometry. Extensive research has been conducted on the cosmological and astrophysical implications of nonminimal matter-geometry coupling [26–29].

The $f(R, L_m)$ gravity theory, which is a more advanced generalization of matter-curvature coupling theories, was recently developed by Harko and Lobo [30]. In this gravity, $f(R, L_m)$ is an arbitrary function of the Ricci scalar R and the matter Lagrangian density L_m . In $f(R, L_m)$ gravity, the motion of test particles is nongeodesic, resulting in the formation of an additional force orthogonal to the four-velocity

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vectors. The equivalence principle, which is severely constrained by solar system testing, is explicitly broken by the $f(R, L_m)$ gravity models. Wang and Liao [31] have recently investigated the energy conditions in $f(R, L_m)$ gravity. Jaybhave et al. [32] examined the cosmic expansion scenario of the universe within the context of $f(R, L_m)$ gravity. Patil et al. [33] used an equation of state (EoS) for strange quark matter (SQM) to characterize the cosmic late-time acceleration of the universe in the $f(R, L_m)$ gravity framework proposed by Harko (2010). Pawde et al. [34] used the framework of $f(R, L_m)$ gravity to study the anisotropic behavior of the universe.

The literature contains several researches on cosmological models that investigate the correlation between the matter composition and observable constraints under various modified gravity frameworks. Recent developments in cosmology have improved our knowledge of the accelerating expansion of the universe, especially via studies of Type Ia supernovae and the cosmic microwave background (CMB). In this regard, Odintsov et al. (2023) examined early dark energy within $f(R)$ gravity models, providing new insights into the influence of dark energy during the early stages of the universe [35]. In the same way, Myrzakulova et al. (2024) looked into observational constraints in $f(R, L_m)$ models and came up with important results about how dark energy and matter interaction [36]. Studies like Sahoo et al. (2017), which examined the implications of $f(R, T)$ gravity for cosmic expansion, enhance these advancements in modified gravity theories [37]. In 2024, Pawar et al. used $f(T)$ gravity to study the flow of matter and radiation in a Bianchi Type-I model that was locally rotationally symmetric. Fitting the Hubble parameter $H(z)$ using 57 observational data points yielded a satisfactory match with the Observational Hubble Dataset (OHD) with the R^2 value of 0.9321 [38]. Finally, the research by Ajmal and Bhatti (2023) have helped us to learn more about how matter and geometry interact in different modified gravity models [39]. Such studies provide the observational foundation and theoretical advancements that support modified gravity models, including $f(R, L_m)$ gravity.

In this study, we used the MSQM distribution for Bianchi type-VI in $f(R, L_m)$ gravity. Many people view the magnetic field as the most significant aspect of the universe. Strong magnetic fields have gained popularity in the scientific community due to their unique qualities related to dense quark matter, neutron star matter, and SQM stability [40–43]. When the electromagnetic scale reaches the order of the nuclear scales, significant changes in the characteristics of odd matter show up, according to the research on quark matter in strong magnetic fields [44, 45].

Research on MSQM under magnetic field conditions is widely accepted. In the modified theory involving MSQM, it was observed that the universe begins with a Big Bang and ends with a Big Rip [46]. The magnetic field is a unique feature of the universe, and it gaining popularity due to its impact on SQM characteristics. The EoS for SQM employing density-independent bag pressure in quark confinement is as follows [47]:

$$p = \frac{1}{3}(\rho - 4\mathcal{B}_c), \quad (1)$$

where $\mathcal{B}_c$ is the bag constant, it is defined as the constant difference in energy densities between the perturbative vacuum and physical systems. The bag constant is measured in MeV/fm^3 and ranges from 60 to 80 MeV/fm^3 [48]. Sahoo et al. [46], Aktaş and Aygün [49], and Solanke et al. [50] recently examined MSQM in the context of $f(R, T)$ gravity.

This work is organized as follows: Section 2 contains a brief review of DE. In Section 3, we derive the basic formalization of $f(R, L_m)$ gravity. The solutions of field equations are determined in Section 4. In Section 5, we discuss the physical model of $f(R, L_m)$ gravity. Physical and kinematical properties are discussed in Section 6. We compare the plot of the Hubble parameter with the Hubble dataset in Section 7. In Section 8, we look at the energy conditions. The observations and discussion of the graphical resolution of parameters are presented in Section 9. Lastly, in Section 10, the conclusion and perspective of our approach are outlined.

2. A BRIEF REVIEW OF DARK ENERGY

Unsolved cosmological riddles persist regarding the nature of the DE component of the universe. There are undoubtedly plenty of applications that are phantom energy, quintessence, k -essence, a remnant of the cosmological constant, Tsallis holographic DE (THDE), and ghost DE (GDE), Rényi holographic DE (RHDE), etc. Since our universe is currently experiencing an accelerated expansion, which has been confirmed by numerous observations including type-I supernovae (SNeIa), the Sloan Digital Sky Survey (SDSS), and the Wilkinson Microwave Anisotropy Probe (WMAP), there has been a lot of interest recently in cosmological models with DE in general relativity (GR) [6, 7, 51–55]. Based on these findings, the researchers have acknowledged DE. To explain the facts, the researchers have explored several possibilities for DE, including the cosmological constant, tachyon, quintessence, phantom, etc. The current study looks at the observational evidence to find the DE part of the universe's EoS parameter ($\omega = p/\rho$), which might or might not remain the same.

Recent calculations of the parameter ω by several authors involved some reasoning that ultimately reduced to a basic parametrization of the dependencies [56–60]. Our understanding of the universe is based on these facts, which show that it is flat and full of undamped forms of energy density. The undamped energy known as DE, which exists under negative pressure, composes roughly 74% of the total energy density. Measurable amounts of matter comprise the remaining 26% of the energy density, consisting of approximately 22% of dark matter (DM) and 4% of baryonic matter.

Thus, one of the most difficult issues in astrophysics and cosmology today is figuring out the nature of DE. Current cosmological measurements show that our universe is expanding faster than a matter-dominated universe, which contradicts the theory of decelerated expansion.

3. BASIC FORMULATION OF $f(R, L_m)$ GRAVITY

The action principle for $f(R, L_m)$ gravity with the matter Lagrangian density L_m and the Ricci scalar R is

$$S = \int f(R, L_m) \sqrt{-g} d^4x, \quad (2)$$

where f is an arbitrary function. The metric tensor g_{ij} and the Ricci tensor R_{ij} are used to define the Ricci scalar R as

$$R = g^{ij} R_{ij}, \quad (3)$$

where the Ricci tensor is given by

$$R_{ij} = \partial_k \Gamma_{ij}^k - \partial_j \Gamma_{ik}^k + \Gamma_{ij}^l \Gamma_{lk}^k + \Gamma_{jl}^k \Gamma_{ik}^l. \quad (4)$$

The components of the well-known Levi-Civita connection are

$$\Gamma_{jk}^i = \frac{1}{2} g^{il} (g_{kl,j} + g_{lj,k} - g_{jk,l}). \quad (5)$$

The field equation for $f(R, L_m)$ gravity, determined by varying the action (2) in the metric tensor g_{ij} , is stated below:

$$\begin{aligned} f_R R_{ij} - \frac{1}{2} (f - f_{L_m} L_m) g_{ij} + (g_{ij} \square - \nabla_i \nabla_j) f_R \\ = \frac{1}{2} L_m T_{ij}, \end{aligned} \quad (6)$$

where $f_R(R, L_m) = \frac{\delta f(R, L_m)}{\delta R}$, $f_{L_m}(R, L_m) = \frac{\delta f(R, L_m)}{\delta L_m}$, $\square = \nabla_i \nabla^i$, and T_{ij} is the energy-momentum tensor expressed as

$$T_{ij} = \frac{-2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} L_m)}{\delta g^{ij}} = g_{ij} L_m - 2 \frac{\partial L_m}{\partial g^{ij}}. \quad (7)$$

We may determine the covariant divergence of the energy-momentum tensor T_{ij} by deriving the explicit expression of field equation (6) in the following manner,

$$\nabla^i T_{ij} = 2 \nabla^i \ln(f_{L_m}) \frac{\partial L_m}{\partial g^{ij}}. \quad (8)$$

Additionally, we can demonstrate the relationship between the matter Lagrangian density L_m , the Ricci scalar R , and the trace T of the stress-energy-momentum-tensor T_{ij} , by contracting the field equation (6) as

$$R f_R + 3 \square f_R - (f - f_{L_m} L_m) = \frac{1}{2} f_{L_m} T, \quad (9)$$

where $\square F = \frac{1}{\sqrt{-g}} \partial_i (\sqrt{-g} g^{ij} \partial_j F)$.

4. METRIC AND EQUATIONS OF MOTION IN $f(R, L_m)$ GRAVITY

We consider the space-time of a general Bianchi type-VI model with the metric [61]

$$\begin{aligned} ds^2 = -dt^2 + A^2 dx^2 + B^2 e^{-2\beta x} dy^2 \\ + C^2 e^{2\beta x} dz^2, \end{aligned} \quad (10)$$

where β is a constant, A, B, C are functions of t .

The Ricci scalar associated with the general Bianchi type-VI space-time is

$$\begin{aligned} R = -2 \left[\frac{\ddot{A}}{A} + \frac{\ddot{B}}{B} + \frac{\ddot{C}}{C} + \frac{\dot{A}\dot{B}}{AB} + \frac{\dot{A}\dot{C}}{AC} \right. \\ \left. + \frac{\dot{B}\dot{C}}{BC} - \frac{\beta^2}{A^2} \right]. \end{aligned} \quad (11)$$

The energy-momentum tensor of the MSQM [62, 63] is taken as

$$\begin{aligned} T_{ij} = (\rho + p + h^2) u_i u_j \\ + \left(\frac{h^2}{2} - p \right) g_{ij} - h_i h_j, \end{aligned} \quad (12)$$

where the four-velocity vector is $u^i = (0, 0, 0, 1)$, fulfilling $u_i u^i = 1$. Additionally, with $h_j u^j = 0$, the magnetic flux h^2 is obtained from the x direction, p is the pressure, and ρ is the energy density.

The field equations (6) take the form

$$\begin{aligned} -f_R \left(\frac{\ddot{A}}{A} + \frac{\dot{A}\dot{B}}{AB} + \frac{\dot{A}\dot{C}}{AC} - \frac{\beta^2}{A^2} \right) - \frac{1}{2} (f - f_{L_m} L_m) \\ - \dot{f}_R \left(\frac{\dot{B}}{B} + \frac{\dot{C}}{C} \right) - \ddot{f}_R = \frac{1}{2} f_{L_m} \left(\frac{h^2}{2} - p \right), \end{aligned} \quad (13)$$

$$\begin{aligned}
& -f_R \left(\frac{\ddot{B}}{B} + \frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}\dot{C}}{BC} \right) - \frac{1}{2}(f - f_{L_m}L_m) \\
& - \dot{f}_R \left(\frac{\dot{A}}{A} + \frac{\dot{B}}{B} + \frac{\dot{C}}{C} - \frac{\beta^2}{A^2} \right) - \ddot{f}_R \\
& = \frac{1}{2}f_{L_m} \left(\frac{h^2}{2} - p \right), \quad (14)
\end{aligned}$$

$$\begin{aligned}
& -f_R \left(\frac{\ddot{C}}{C} + \frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}\dot{C}}{BC} \right) - \frac{1}{2}(f - f_{L_m}L_m) \\
& - \dot{f}_R \left(\frac{\dot{A}}{A} + \frac{\dot{B}}{B} + \frac{\dot{C}}{C} - \frac{\beta^2}{A^2} \right) - \ddot{f}_R \\
& = \frac{1}{2}f_{L_m} \left(\frac{h^2}{2} - p \right), \quad (15)
\end{aligned}$$

$$\begin{aligned}
& -f_R \left(\frac{\ddot{A}}{A} + \frac{\ddot{B}}{B} + \frac{\ddot{C}}{C} \right) - \frac{1}{2}(f - f_{L_m}L_m) \\
& - \dot{f}_R \left(\frac{\dot{A}}{A} + \frac{\dot{B}}{B} + \frac{\dot{C}}{C} \right) = \frac{1}{2}f_{L_m} \left(\frac{3h^2}{2} + \rho \right), \quad (16) \\
& f_R \left(\frac{\dot{B}}{B} - \frac{\dot{C}}{C} \right) \beta = 0. \quad (17)
\end{aligned}$$

From Eq. (17) we get

$$B = c_1 C, \quad (18)$$

where c_1 is an integration constant. Without loss of generality, we can consider $c_1 = 1$ for simplicity, hence we get

$$B = C. \quad (19)$$

5. COSMOLOGICAL SOLUTIONS FOR A PHYSICAL MODEL OF $f(R, L_m)$ GRAVITY

In this research, we concentrate on the cosmological implications of $f(R, L_m)$ gravity, with a relationship between R and L_m being [64]

$$f(R, L_m) = \frac{R}{2} + \lambda R^2 + \alpha L_m. \quad (20)$$

The assumed model $f(R, L_m)$ reduces to GR for the values $\lambda = 0$ and $\alpha = 1$.

We have examined a dusty universe scenario for this specific functional form of $f(R, L_m)$ gravity, where we apply $L_m = \rho$ [29, 33, 34, 65]. Using Eq. (19) in the field equations (13)–(16), we get

$$\left(\frac{1}{2} + 2\lambda R \right) \left(\frac{\ddot{A}}{A} + 2\frac{\dot{A}\dot{B}}{AB} - \frac{\beta^2}{A^2} \right)$$

$$\begin{aligned}
& + \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \\
& \times \left[\frac{1}{2} + \lambda \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \right] \\
& = \frac{1}{2}\alpha \left(\frac{h^2}{2} - p \right), \quad (21)
\end{aligned}$$

$$\begin{aligned}
& - \left(\frac{1}{2} + 2\lambda R \right) \left(\frac{\ddot{B}}{B} + \frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} \right) \\
& + \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \\
& \times \left[\frac{1}{2} + \lambda \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \right] \\
& = \frac{1}{2}\alpha \left(\frac{h^2}{2} - p \right), \quad (22)
\end{aligned}$$

$$\begin{aligned}
& - \left(\frac{1}{2} + 2\lambda R \right) \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} \right) \\
& + \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \\
& \times \left[\frac{1}{2} + \lambda \left(\frac{\ddot{A}}{A} + 2\frac{\ddot{B}}{B} + 2\frac{\dot{A}\dot{B}}{AB} + \frac{\dot{B}^2}{B^2} - \frac{\beta^2}{A^2} \right) \right] \\
& = \frac{1}{2}\alpha \left(\frac{3h^2}{2} + \rho \right). \quad (23)
\end{aligned}$$

Equations (21)–(23) form a system of three equations with five unknowns A, B, p, ρ , and h^2 .

Observations in cosmology focus on physical characteristics like the spatial volume (V), the Hubble parameter (H), the expansion scalar (θ), the anisotropy parameter (A_m), the shear scalar (σ^2), and the deceleration parameter (q). They are defined as

$$V = a^3 = AB^2, \quad (24)$$

where a is the average scale factor,

$$H = \dot{a}/a, \quad (25)$$

$$\theta = 3H, \quad (26)$$

$$A_m = \frac{1}{3} \sum_{i=1}^3 \frac{(H_i - H)}{H}, \quad (27)$$

where H_i ($i = 1, 2, 3$) are the directional Hubble parameters along the x, y , and z axes,

$$\sigma^2 = \frac{3}{2} A_m H^2, \quad (28)$$

$$q = -\frac{a\ddot{a}}{\dot{a}^2}. \quad (29)$$

The expansion rate of the universe is measured by the deceleration parameter. The sign of q indicates how the universe is expanding. Thus, $q > 0$ denotes deflation, and $q < 0$ indicates inflation. Expansion with a constant velocity occurs at $q = 0$. Berman [66] and Berman and Gomide [67] proposed a law of variation for the Hubble parameter in the context of FLRW (Friedmann–Lemaître–Robertson–Walker) space–time in GR. This led to a constant deceleration parameter ($q = m - 1, m \geq 0$). Following the discovery of the present acceleration of the universe, several writers have examined cosmological models utilizing Berman's law in the context of DE. This is because the deceleration parameter in Berman's law can have values $q \geq -1$ and $-1 \leq q < 0$ corresponding to the accelerating expansion. This study focuses on the linearly varying deceleration parameter [68], which is given by

$$q = -kt + m - 1, \quad (30)$$

where $k \geq 0$ and $m \geq 0$ are constants, $k = 0$, and Eq. (30) reduces to Berman's law [65, 66]. This law allows for a generalization of constant deceleration parameter-derived cosmological solutions. Our linearly varying deceleration parameter approach leads to a universe undergoing super-exponential expansion unless $k = 0$, which is one of the conceivable outcomes of the universe based on the cosmological evidence.

From Eqs. (29) and (30), we get the value of the average scale factor as

$$a = c_2 \exp \left[\frac{2}{\sqrt{m^2 - 2c_3k}} \operatorname{arctanh} \left(\frac{kt - m}{\sqrt{m^2 - 2c_3k}} \right) \right] \quad \text{for } k > 0 \text{ and } m \geq 0, \quad (31)$$

where c_2, c_3 are integration constants. To simplify, we explore the solution for $k > 0$ and $m > 0$, omitting the integration constant c_3 by $c_3 = 0$. By doing so, we also set the initial time of the universe to $t_i = 0$. The answer for $k > 0$ and $m > 0$ is chosen not only for simplicity but also for conformity with the known world. If $k > 0$, it signifies an increase in the acceleration. Since $t_i = 0$ and $k > 0$, the only way to increase the deceleration parameter beyond -1 is by setting $m > 0$. Now, by choosing $m > 1$, that is, m is significantly larger than k , under this consideration, Eq. (31) reduces to

$$a = c_2 \exp \left[\frac{2}{m} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right]. \quad (32)$$

Now, using the above equation, we get the Hubble parameter as

$$H = \frac{\dot{a}}{a} = \frac{-2}{t(kt - 2m)}. \quad (33)$$

Using Eqs. (24) and (32), we have

$$\left(c_2 \exp \left[\frac{2}{m} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right] \right)^3 = AB^2. \quad (34)$$

The expansion scalar θ is proportional to the shear scalar σ^2 , which is a physical condition we employ to solve the aforementioned equations. This condition leads to [69]

$$B = A^n, \quad (35)$$

where n is an arbitrary constant.

Using Eqs. (34) and (35), we obtain

$$A = c_2 \exp \left[\frac{6}{m(2n+1)} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right], \quad (36)$$

$$B = c_2 \exp \left[\frac{6n}{m(2n+1)} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right]. \quad (37)$$

Therefore, a Bianchi type-VI space–time is described by the line element (10), corresponding to Eqs. (36) and (37), which can be expressed as follows:

$$\begin{aligned} ds^2 = & -dt^2 \\ & + c_2 \exp \left[\frac{12}{m(2n+1)} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right] dx^2 \\ & + c_2 \exp \left[\frac{12n}{m(2n+1)} \operatorname{arctanh} \left(\frac{kt}{m} - 1 \right) \right] \\ & \times \left(e^{-2\beta x} dy^2 + e^{2\beta x} dz^2 \right). \end{aligned} \quad (38)$$

6. PHYSICAL AND KINEMATIC PROPERTIES OF THE MODEL

The redshift is a highly significant and often contentious phenomenon regarding cosmology and astronomy. When electromagnetic radiation from an object is directed towards the less energetic regions of the spectrum, a phenomenon known as the shift in electromagnetic radiation occurs. Utilizing this phenomenon, we get insight into aspects of our galaxy and even the whole universe. The relation between the average scale factor a and the redshift z [70] is

$$a = \frac{1}{1+z}. \quad (39)$$

In this section, we estimate and determine significant cosmological quantities in terms of the redshift. These computations aid our understanding of the expansion of the universe, the energy distribution, and the effects of many components at various redshifts.

The key cosmological parameters, the deceleration parameter q , the Hubble parameter H , the anisotropy parameter A_m , the expansion scalar θ , and the shear scalar σ^2 in terms of the redshift are calculated as

$$q = m \tanh \left[c_2 \frac{m}{2} \ln(1+z) \right] - 1, \quad (40)$$

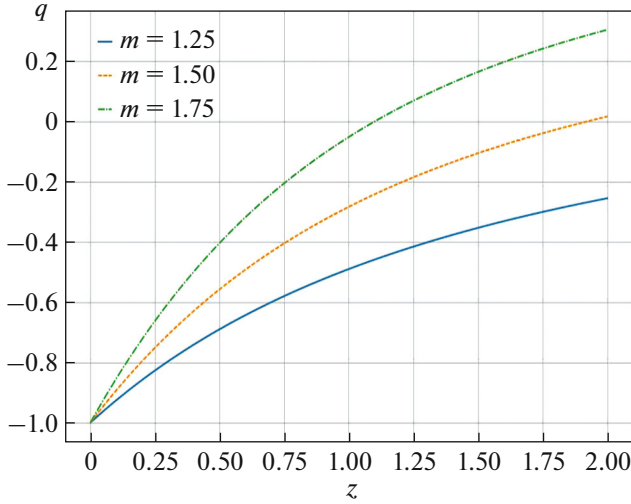


Fig. 1. Variation of the deceleration parameter q versus redshift z for the particular choice of constant $c_2 = 1$.

$$H = \frac{2k}{m^2} \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right], \quad (41)$$

$$A_m = -\frac{m^2}{6k \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right]} \left[\frac{3(1+n)}{(2n+1)(1+z)} + \frac{4k}{m^2} \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right] \right], \quad (42)$$

$$\theta = \frac{6k}{m^2} \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right], \quad (43)$$

$$\sigma^2 = -\frac{k}{m^2} \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right] \left[\frac{3(1+n)}{(2n+1)(1+z)} + \frac{4k}{m^2} \cosh^2 \left[c_2 \frac{m}{2} \ln(1+z) \right] \right]. \quad (44)$$

Solving Eqs. (21), (22), and (23) and using Eqs. (1), (36), and (37), we obtain the energy density ρ , the pressure p , and the magnetic flux h^2 in terms of the redshift as

$$\rho = \frac{3}{\alpha} \left\{ \frac{1}{2} + 2\lambda \left[\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] \right\} \left[\frac{2(3+4n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] - \frac{2}{\alpha} \left\{ \frac{1}{2} + \lambda \left[\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] \right\} \left[\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] + 2\mathcal{B}_c, \quad (45)$$

$$p = \frac{1}{3} \left\{ \frac{3}{\alpha} \left[\frac{1}{2} + 2\lambda \left(\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right) \right] \left[\frac{2(3+4n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] - \frac{2}{\alpha} \left[\frac{1}{2} + \lambda \left(\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right) \right] \left[\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] - 2\mathcal{B}_c \right\}, \quad (46)$$

$$h^2 = \frac{2}{3} \left\{ \frac{3}{\alpha} \left[\frac{1}{2} + 2\lambda \left(\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right) \right] \left[\frac{-16(3+5n)}{(2n+1)^2(1+z)^2} \right] + \frac{4}{\alpha} \left[\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right] \times \left[\frac{1}{2} + 2\lambda \left(\frac{-6(4+(13n+8)n)}{(2n+1)^2(1+z)^2} - 2c_2\beta^2(1+z)^{\frac{3}{2n+1}} \right) \right] - 2\mathcal{B}_c \right\}. \quad (47)$$

An important idea in cosmology is the EoS parameter ω , which specifies the connection between the pressure p and the energy density ρ of the universe as

$$\rho\omega = p. \quad (48)$$

This parameter contributes to describing the nature of several components in the universe: dust (non-

relativistic matter) ($\omega = 0$), radiation ($\omega = 1/3$), and vacuum energy ($\omega = -1$) that is mathematically equivalent to a cosmological constant Λ . It is commonly recognized that $\omega < -1/3$ is typically considered when discussing DE since it causes an accelerated expansion. Several scalar field models have been suggested, including quintessence ($-1 \leq$

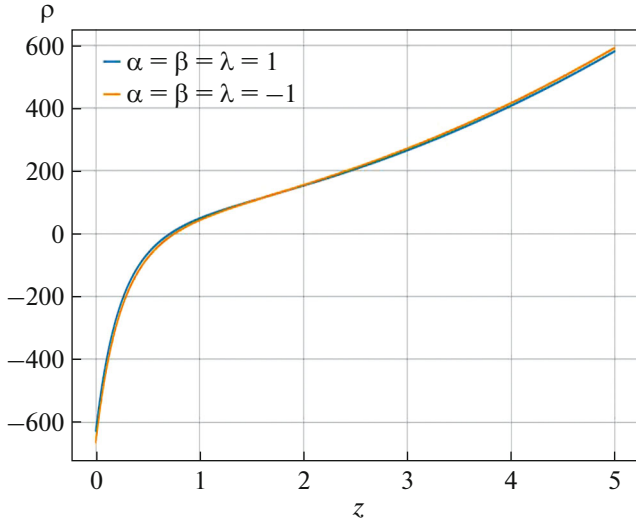


Fig. 2. Variation of the energy density ρ versus redshift z for the particular choice of the constants $n = c_2 = 1$, $B_c = 0.5$.

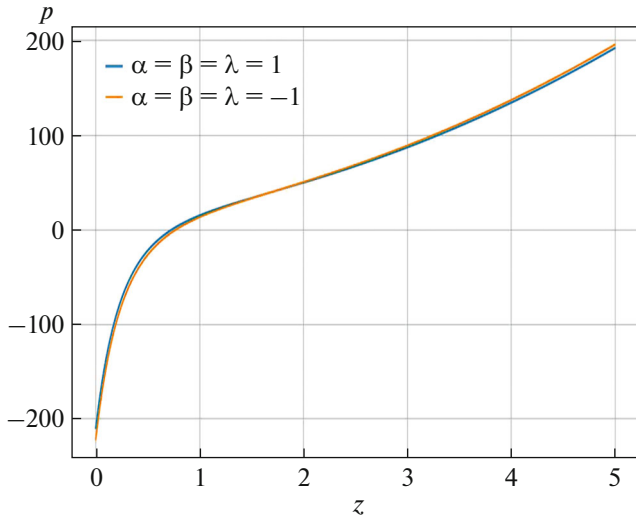


Fig. 3. Variation of the pressure p versus redshift z for the particular choice of constants $n = c_2 = 1$, $B_c = 0.5$.

$\omega \leq 1$), phantom ($\omega \leq -1$), and quintom that can change across the cosmological constant boundary ($\omega = -1$). These have been put forward along with the constant EoS parameter with $\omega < -1/3$, especially in the context of DE. Such models can be described by a time-dependent ω that can evolve below $-1/3$.

The EoS parameter for our model is directly obtained as a ratio of the expressions (45) and (46).

7. THE HUBBLE DATASETS

Measurements of dt yield the Hubble parameter, which is independent of the model, as the Hubble

parameter is also related to the differential redshift as

$$H(z) = -\frac{1}{(1+z)} \frac{dz}{dt},$$

where dz is acquired from spectroscopic surveys. Actually, there are two common approaches to determining the Hubble parameter values $H(z)$ at a given redshift: a differential age (DA) and the line-of-sight baryonic acoustic oscillation (BAO) datasets. The 57 newest $H(z)$ data points in the redshift range of $0.070 \leq z \leq 2.36$ provide observational constraints on the parameter $H_0 = \epsilon(\lambda + 1)$. The DA technique was used to collect 31 points, and the BAO technique was used to collect 26 points. A list of 57 $H(z)$ values in the redshift range $0.070 \leq z \leq 2.36$ has been proposed in [63–65] (see Table 1). We found the best-fit curve of $H(z)$ with 57 data values shown in Table 1, using the R^2 test:

$$R^2 = 1 - \frac{\sum_{i=1}^{57} [(H_i)_{\text{obs}} - (H_i)_{\text{th}}]^2}{\sum_{i=1}^{57} [(H_i)_{\text{obs}} - (H_i)_{\text{mean}}]^2}. \quad (49)$$

The R^2 test examines how much of the universe in the dependent variable can be explained by the independent variable. Regarding the values of the model parameters H_0 , δ , and λ in relation to the observatory Hubble dataset (OHD), $R^2 = 1$ denotes an exact fit. To determine the optimal values of H_0 , δ , and λ , we limit the parametric spaces to $-1 < z$ and $\epsilon \neq 0$. For $H_0 = 67.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$, the error bars show the mean point and how far it is from the mean across 57 points of the Hubble dataset,

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}.$$

Here, Ω_m and Ω_Λ are the density parameters of dark matter and DE, respectively. For approximate values of $H_0 = 64.49^{+0.33}_{-0.32}$, $\delta = 1.54^{+0.02}_{-0.02}$, $\lambda = 1.14^{+0.068}_{-0.077}$, we obtain the best-fit plot with a maximum of $R^2 = 0.9321$ and a root mean square error (RMSE) of 11.071.

8. ENERGY CONDITIONS

The energy conditions are fundamental in GR to ensure the physical viability of cosmological models. They limit the energy density and pressure to avert nonphysical situations, such as negative energy densities or superluminal energy transmission. Within the framework of modified gravity theories, the energy conditions serve as significant tests to confirm whether the assumptions regarding energy and matter are compatible with the observable development of the universe. In general, the energy conditions include:

Table 1. The Hubble datasets

z	$H(z)$	σ_H	Ref.	z	$H(z)$	σ_H	Ref.
0.070	69	19.6	[73]	1.750	202	40	[74]
0.90	69	12	[74]	1.965	186.5	50.4	[79]
0.120	68.6	26.2	[73]	0.24	79.69	2.99	[80]
0.170	83	8	[74]	0.30	81.7	6.22	[81]
0.1791	75	4	[75]	0.31	78.18	4.74	[82]
0.1993	75	5	[75]	0.34	83.8	3.66	[80]
0.200	72.9	29.6	[76]	0.35	82.7	9.1	[83]
0.270	77	14	[74]	0.36	79.94	3.38	[82]
0.280	88.8	36.6	[76]	0.38	81.5	1.9	[84]
0.3519	83	14	[75]	0.40	82.04	2.03	[82]
0.3802	83	13.5	[77]	0.43	86.45	3.97	[80]
0.400	95	17	[74]	0.44	82.6	7.8	[85]
0.4004	77	10.2	[77]	0.44	84.84	1.83	[82]
0.4247	87.1	11.2	[77]	0.48	87.79	2.03	[82]
0.4497	92.8	12.9	[77]	0.51	90.4	1.9	[84]
0.470	89	34	[78]	0.52	94.35	2.64	[82]
0.4783	80.9	9	[77]	0.56	93.34	2.3	[82]
0.480	97	62	[73]	0.57	87.6	7.8	[86]
0.593	104	13	[75]	0.57	96.8	3.4	[87]
0.6797	92	8	[75]	0.59	98.48	3.18	[82]
0.7812	105	12	[75]	0.60	87.9	6.1	[85]
0.8754	125	17	[75]	0.61	97.3	2.1	[84]
0.880	90	40	[73]	0.64	98.82	2.98	[82]
0.900	117	23	[74]	0.73	97.3	7.0	[85]
1.037	154	20	[75]	2.30	224	8.6	[88]
1.300	168	17	[74]	2.33	224	8	[89]
1.363	160	33.6	[79]	2.34	222	8.5	[90]
1.430	177	18	[74]	2.36	226	9.3	[91]
1.530	140	14	[74]				

- The Strong energy condition (SEC): the inequality $\rho + 3p \geq 0$, gravity is always attractive.
- The Dominant energy condition (DEC): The matter-energy density will be positive and propagate causally when an observer measures, resulting in $\rho \geq |p|, \rho \geq 0$. The DEC suggests that sound energy will not travel faster than light.
- The Weak energy condition (WEC): Any time-like observer measuring matter-energy density should find that $\rho \geq 0, \rho + p \geq 0$. The WEC suggests that there should not be a negative energy density.
- The Null energy condition (NEC) is provided by $\rho + p \geq 0$ for a perfect fluid energy-momentum tensor.

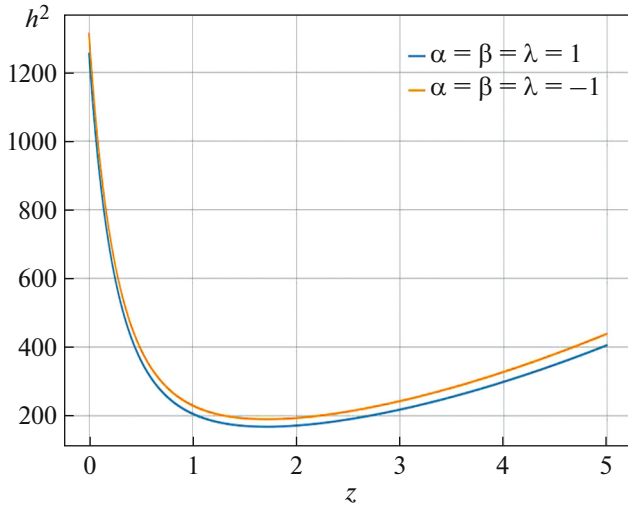


Fig. 4. Variation of the magnetic flux h^2 versus redshift z for the particular choice of constants $n = c_2 = 1$, $B_c = 0.5$.

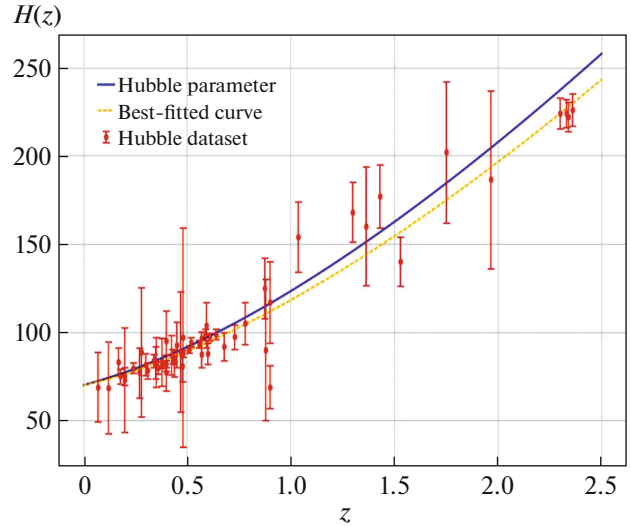


Fig. 6. The Hubble parameter H versus redshift z along with the Hubble dataset given in Table 1.

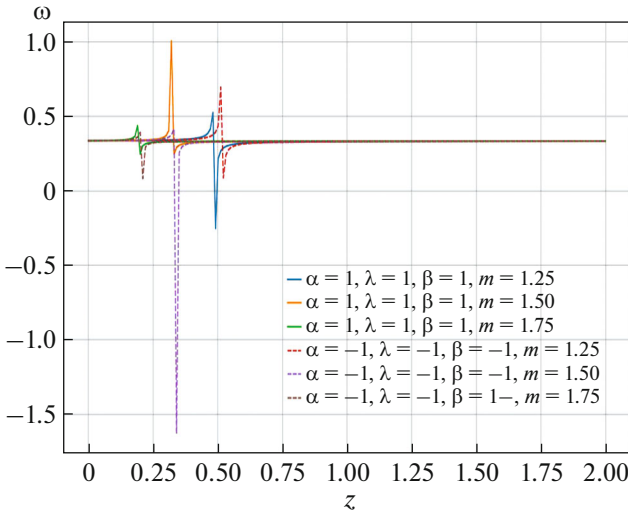


Fig. 5. Variation of the EoS parameter ω versus redshift z for the particular choice of constants $n = c_2 = 1$, $B_c = 0.5$.

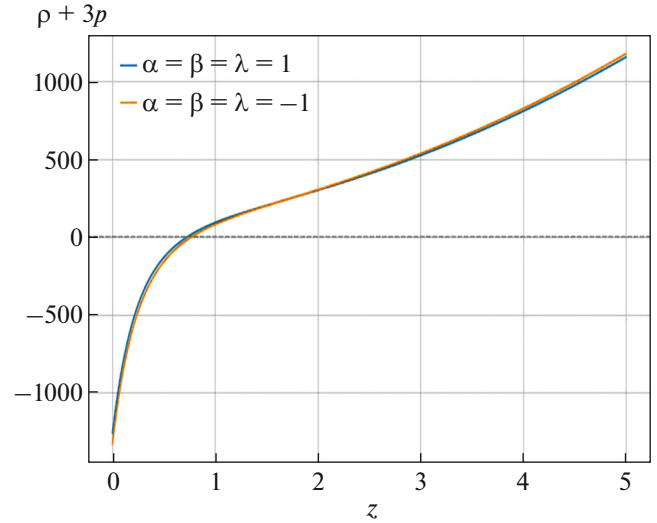


Fig. 7. Variation of $\rho + 3p$ versus redshift z .

Using Eqs. (45) and (46), we obtain the energy conditions for our analysis.

9. OBSERVATIONS AND DISCUSSION

Several notable observations may be made within the context of the shown figures.

- The deceleration parameter q indicates whether the universe is accelerating or decelerating in its expansion. At $z = 0$, the deceleration parameter is negative for all values of m , which suggests that the universe is currently undergoing accelerated expansion. Figure 1

shows that the expansion of the universe has gone through several phases, starting with an early deceleration period and ending with the current accelerated phase. Higher values of m result in an earlier transition and slower current acceleration, and in general, the parameter m controls the time and impact of these phases.

- The Hubble parameter provides vital information on the rate of expansion of the universe throughout cosmic epochs and its dynamic evolution vs. redshift z . Equation (41) presents the Hubble parameter along with the redshift. The model predicts a finite Hubble parameter at $z = 0$, which is consistent with the current

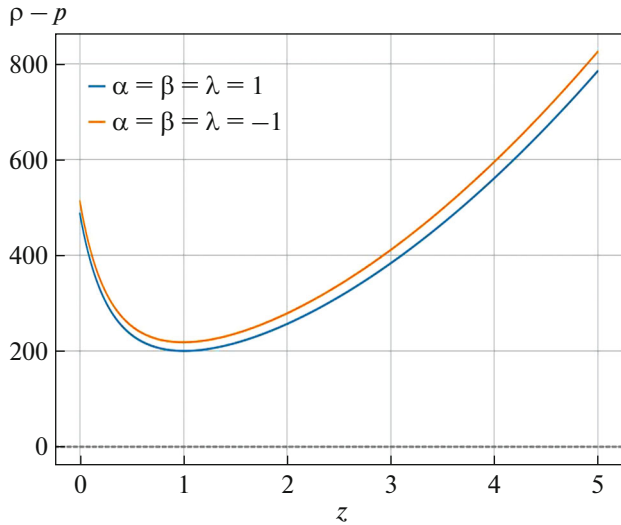


Fig. 8. Variation of $\rho - p$ versus redshift z .

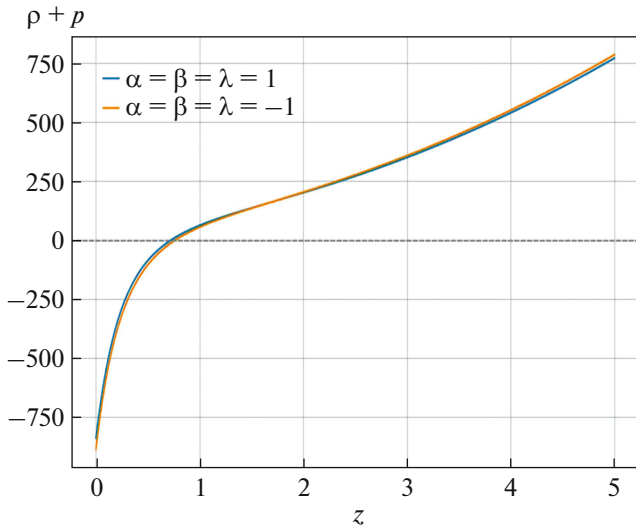


Fig. 9. Variation of $\rho + p$ versus redshift z .

observations of the expansion rate. At high redshifts ($z \rightarrow \infty$), the Hubble parameter rises quickly, pointing to a time of faster expansion that may be connected with the start of inflation. The model predicts an infinite expansion rate as $z \rightarrow -1$, which indicates the possibility of a “Big Rip” scenario, in which the acceleration is not under control. The model requires a nonzero spatial curvature ($k \neq 0$), which may present challenges when compared to observations that favor a flat universe.

- Figure 2 shows variation of the energy density ρ versus redshift z in the changing dynamics of the universe, focusing on the transition from a DE-dominated phase to a matter/radiation-

dominated phase. The specific parameters α, β, λ affect the speed of this transition but do not change the overall trend. This aligns with our current understanding of the history of expansion in the universe.

- Figure 3 shows that at $z = 0$ the pressure p becomes negative, and the universe is dominated by DE, causing its accelerated expansion. As z increases, the transition to positive pressure indicates a shift to matter domination. This aligns with our understanding of the history of the universe, where matter dominated after the radiation era until DE becomes significant again.
- The evolution of the magnetic flux h^2 throughout the history is depicted in Fig. 4, with a time of weakening in between and a stronger presence in the early universe and the current era. The small variations in the two-parameter sets highlight how sensitive the development of the magnetic flux is to the particular values of the cosmological parameters α, β, λ . Understanding this behavior is essential for comprehending the function of magnetic fields in the construction and evolution of cosmic structures, as well as their significance in the cosmic history.
- The impact of varying redshift and cosmological factors on the EoS ω is seen in Fig. 5. The steady curve for higher values of m implies a more constant expansion, whereas the spikes in ω at certain values of m reflect a rapid transition events in the history of the universe expansion.
- We found that the best-fitted curve closely matched the data points when we compare the plot of the Hubble parameter with the Hubble dataset, as shown in Fig. 6. The plot illustrates the history of the universe, which is characterized by an increase in the Hubble parameter as the redshift increases. This increase reflects the evolution from an early universe dominated by radiation to an age dominated by matter, and then eventually to a dark energy-dominated, accelerating expansion phase.
- The adoption of energy conditions in this research is crucial for verifying that the solutions obtained in the $f(R, Lm)$ gravity framework have physical significance. The energy conditions examine whether the energy density and pressure in the universe, influenced by MSQM, conform to anticipated patterns in

the cosmic development. Figures 7, 8, and 9 illustrate how the energy conditions, such as $\rho + 3p$, $\rho - p$, and $\rho + p$ evolve with redshift. The plots indicate that the chosen parameters (α, β, λ) exert a considerable long-term influence on the correlation between pressure and energy density, particularly at higher redshifts. This indicates a dynamic interplay that influences the expansion of the universe, shaped by these model parameters. The graphic representations validate that the energy conditions align with the observable cosmic phenomena, reinforcing the credibility of the model and its predictions regarding the accelerated expansion of the universe.

- The parameters n , c_2 , and $\mathcal{B}_c$ are crucial in determining the redshift-dependent values, including the Hubble parameter, energy density, and pressure. These constants have physical significance and are not arbitrary. To illustrate, n controls the expansion rate of the universe through its effect on the anisotropy, while c_2 controls the scale factor. The bag constant for quark matter $\mathcal{B}_c$ affects the pressure and energy density, particularly for MSQM applications. Contrasting the predictions of the model with the empirical data, such as the Hubble dataset, helps to determine the values of these parameters. This alignment illustrates that n , c_2 , and $\mathcal{B}_c$ may be optimized to correspond with the empirical data, hence improving the accuracy of the model. Given their close relationship to measured cosmic quantities, these parameters are essential for understanding the history of expansion and transitional stages of the universe. Their involvement in the model enhances its relevance in the investigation of modified gravity theories, such as $f(R, L_m)$ gravity.

10. CONCLUDING REMARKS

In this paper, we investigated a Bianchi type-VI cosmological model $f(R, L_m)$ gravity with MSQM. The $f(R, L_m)$ gravity considers the Ricci scalar R and the Lagrangian matter L_m as key components. We looked at the nonlinear functional forms, which are given by $f(R, L_m) = R/2 + \lambda R^2 + \alpha L_m$, where λ and α are free model parameters. The study looked at many cosmological parameters, including the deceleration parameter, the Hubble parameter, energy conditions, and the EoS parameter. We compared these to the redshift z and used 2D graphs to illustrate the relationship between the two. In cosmology, redshift describes the phenomenon wherein light from

far-off galaxies or astronomical objects seems to shift toward the red end of the electromagnetic spectrum.

It shows that the deceleration parameter changes to $q = -1$ at $z = 0$ according to Eq. (40). This means that the universe is growing and speeding up at an exponential rate, which fits with the DE Λ CDM model and points that it will keep growing in the future. The Hubble parameter H and the expansion scalar θ , as given by Eqs. (41) and (43), at $z = 0$ characterize the current expansion rate, and both are determined by the parameters $k = -1, 1$, $m = 1.25, 1.5, 1.75$, $c_2 = 1$. By analyzing these parameters, we can infer how different models of the universe behave at present. Equations (42) and (44) provide the anisotropy parameter (A_m) and the shear scalar (σ^2) at $z = 0$ and offer insightful information about the cosmological model, such as how certain parameters (k , m , c_2) affect the expansion of the universe and other aspects of its characteristics. At $z = 0$, the ratio $\sigma^2/\theta^2 \neq 0$, it implies that this model does not approach isotropy [92].

The energy density ρ in Eq. (45) and the pressure p in Eq. (46) have the same behavior. Also, in the present work, we have focused on the magnetic flux h^2 given in Eq. (47). The EoS of DE plays a crucial role in distinguishing the stages of rapid expansion in the universe. We saw the value of $\omega = p/\rho$. Based on the energy density and pressure values (Eqs. (45) and (46)), we found that ω behaves in a complicated way, as shown in Fig. 6. Our present understanding of cosmology is validated by the fulfilment of the energy conditions related to the redshift. The criteria have been extensively explored and validated in several scenarios, including DE, DM, and cosmic expansion, with consistent findings [33, 34, 93–95].

We have gained new insights into the evolution of the universe, particularly its transition from decelerated to accelerated expansion. Our results, which include how the Hubble parameter, the deceleration parameter, and the energy conditions change over time, show that modified gravity theories are important for understanding how the universe is expanding so quickly right now. The study enhances the current research by highlighting the intricate relationship between magnetic fields, dark energy, and strange quark matter. It demonstrates that these components can be effectively modeled within the $f(R, L_m)$ gravity framework. Anisotropic cosmic expansion is not frequently investigated in conventional $f(R)$ or $f(T)$ gravity models; however, the inclusion of MSQM introduces a novel dimension to this realm. Additionally, the practical relevance of $f(R, L_m)$ gravity theories is improved by the comparison predictions of our model with observational Hubble data. Although

refinements in parameter fitting are necessary, our approach demonstrates the potential of this framework to bridge the gap between theoretical predictions and empirical observations.

All things considered, this study establishes a groundwork for future studies in modified gravity, implying that more complex models containing other types of dark energy and matter may offer a more profound understanding of the universe's accelerated expansion. Our findings not only support the viability of $f(R, L_m)$ gravity in modern cosmological research but also open up avenues for more detailed studies of exotic matter and its role in the cosmic evolution.

ACKNOWLEDGMENTS

We extend our deepest gratitude to the esteemed referee and the editor for their invaluable suggestions, which have greatly enhanced the quality of our research and its presentation.

FUNDING

This work was supported by ongoing institutional funding. No additional grants to carry out or direct this particular research were obtained.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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